

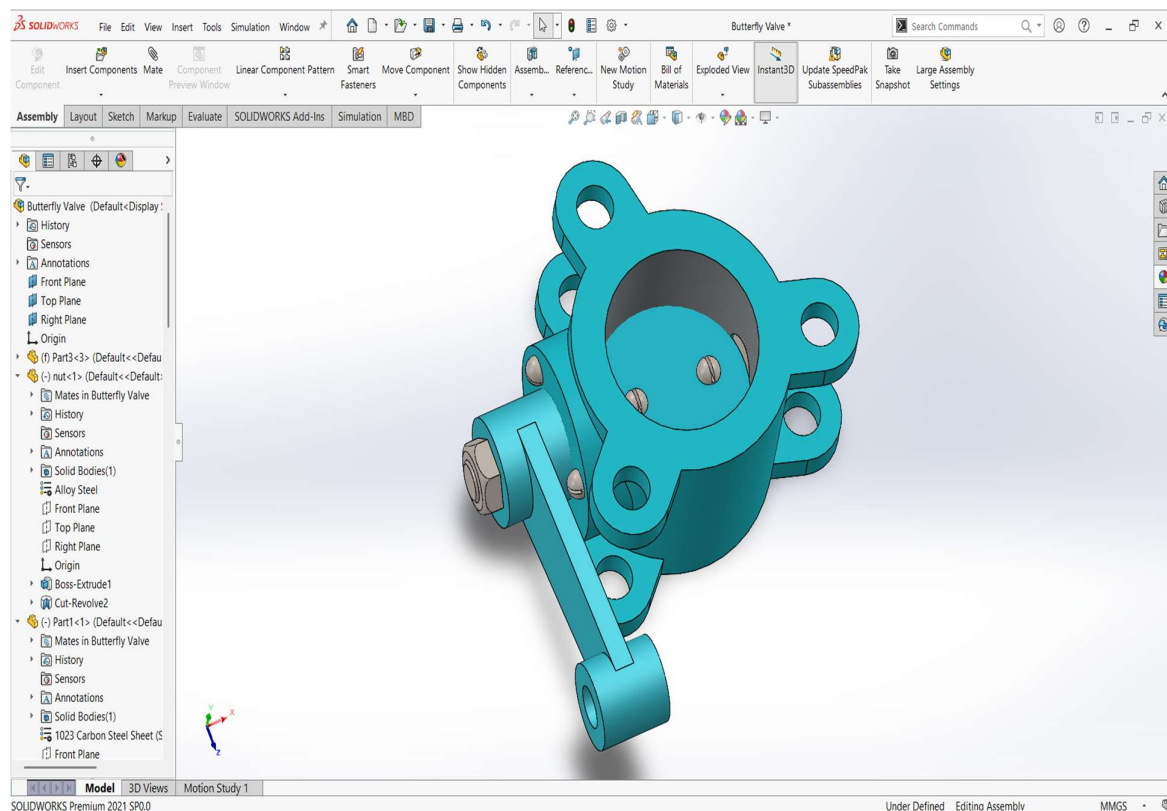
BUTTERFLY VALVE DESIGN AND ASSEMBLY

Designed by: Govind Goyal

Software: SOLIDWORKS 2021 Premium

Project Type: CAD Design | Assembly | Engineering Drawing |
Motion Study

Domain: Mechanical Design Engineering



1. Project Overview

Objective

The objective of this project was to design and assemble a complete butterfly valve using **SOLIDWORKS 2021 Premium** while following standard mechanical design practices. The project focuses on developing practical skills in 3D CAD modeling, assembly creation, engineering drawings, exploded views, motion studies, and design documentation.

This project was completed as part of my personal design engineering portfolio to strengthen my understanding of product development and prepare for a career in mechanical design engineering.

2. About Butterfly Valve

A butterfly valve is a quarter-turn flow control valve used to regulate or isolate the flow of fluids in pipelines. It consists of a rotating circular disc mounted on a shaft. When the lever rotates by 90°, the disc opens or closes the flow passage.

Butterfly valves are widely used because they are lightweight, compact, easy to operate, and require less maintenance than many other valve types.

3. Applications

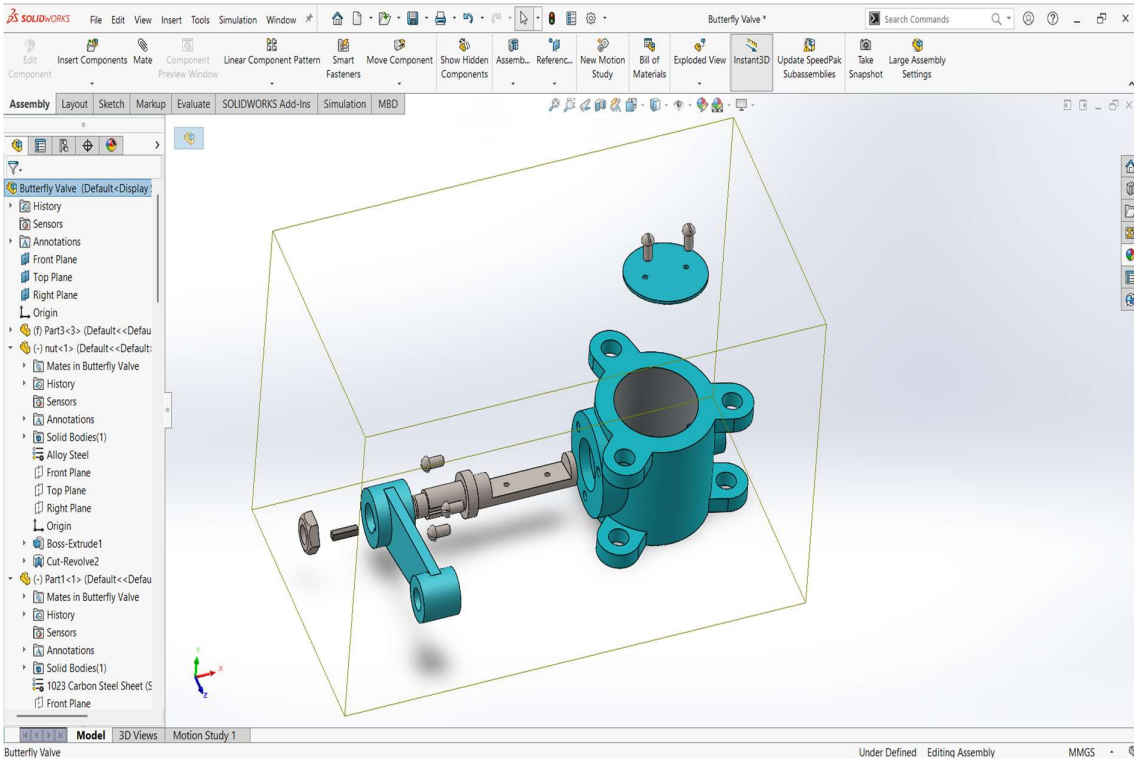
The butterfly valve is commonly used in:

- Water supply systems
- Oil and gas pipelines
- Chemical processing plants
- Food and beverage industries
- HVAC systems
- Fire protection systems
- Pharmaceutical industries
- Power plants

4. Components Used

The assembly consists of the following components:

PART	Quantity
Housing	1
Disc	1
Shaft	1
Lever	1
Pin	1
Nut	1
Screws	5



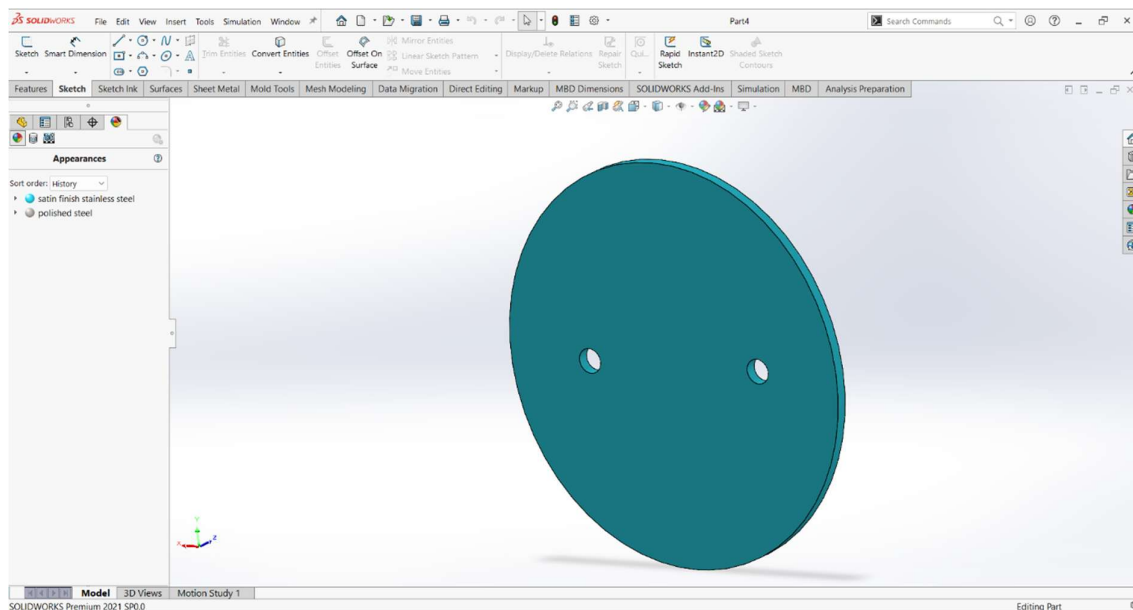
5. Material Selection

Component	Material	Reason
Housing	Grey Cast Iron	High strength, excellent casting properties, vibration damping
Disc	Stainless Steel 316	Corrosion resistance and durability
Shaft	Stainless Steel 304	Good torsional strength and corrosion resistance
Lever	Mild Steel	Easy manufacturing and sufficient strength
Pin	Alloy Steel	High wear resistance
Nut	Alloy Steel	High fastening strength
Screws	Alloy Steel	Reliable clamping force

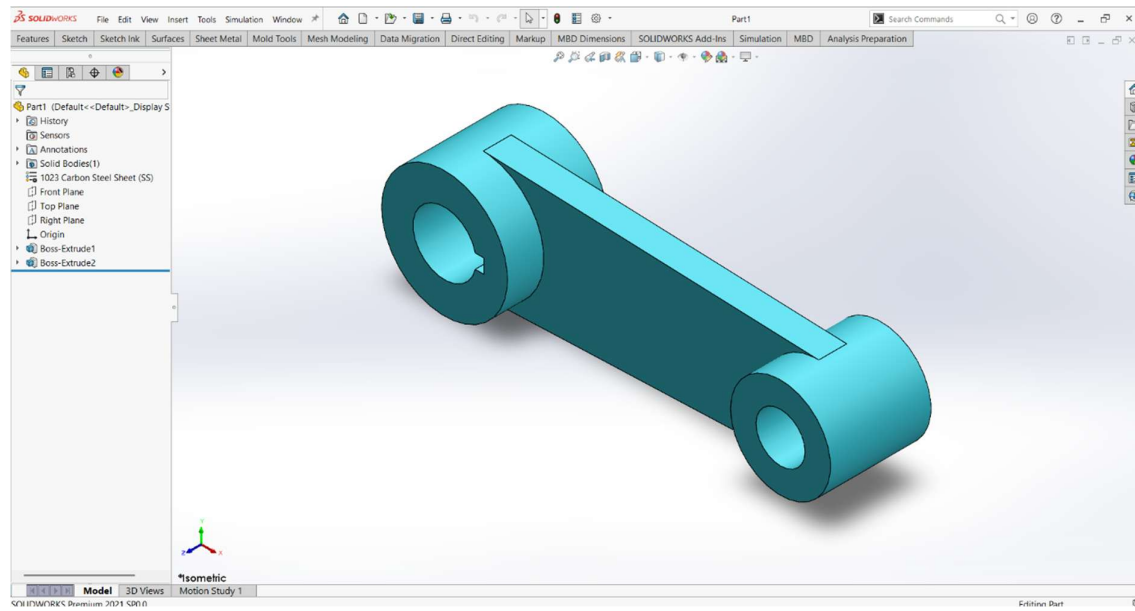
6. CAD Modeling

Each part was designed separately using standard SOLIDWORKS features such as:

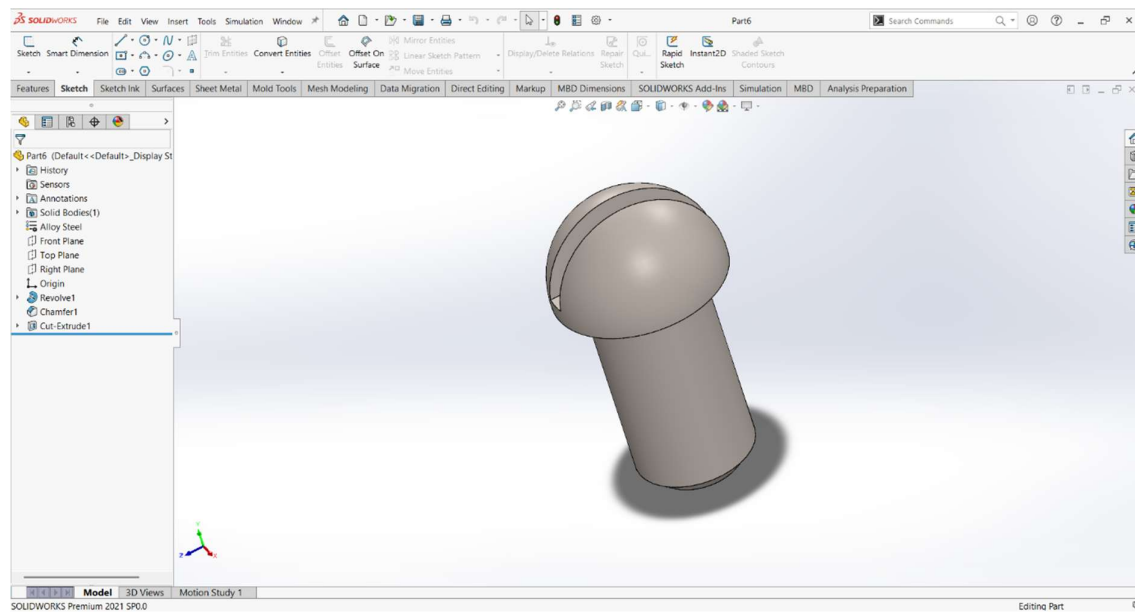
- Extrude Boss/Base



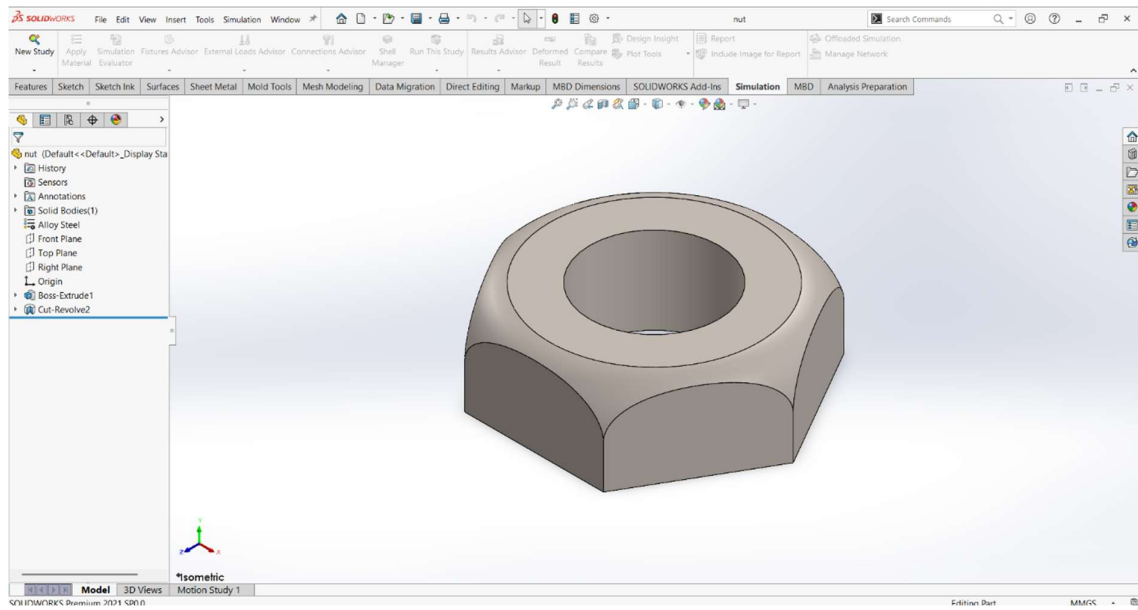
• Extrude Cut



• Revolve

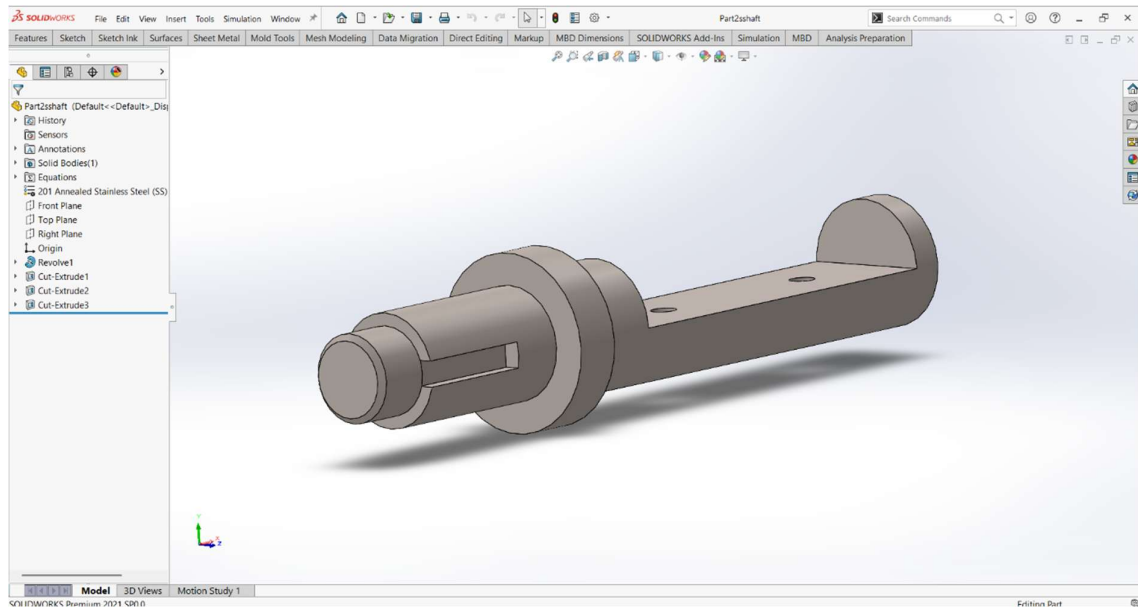


• Fillet

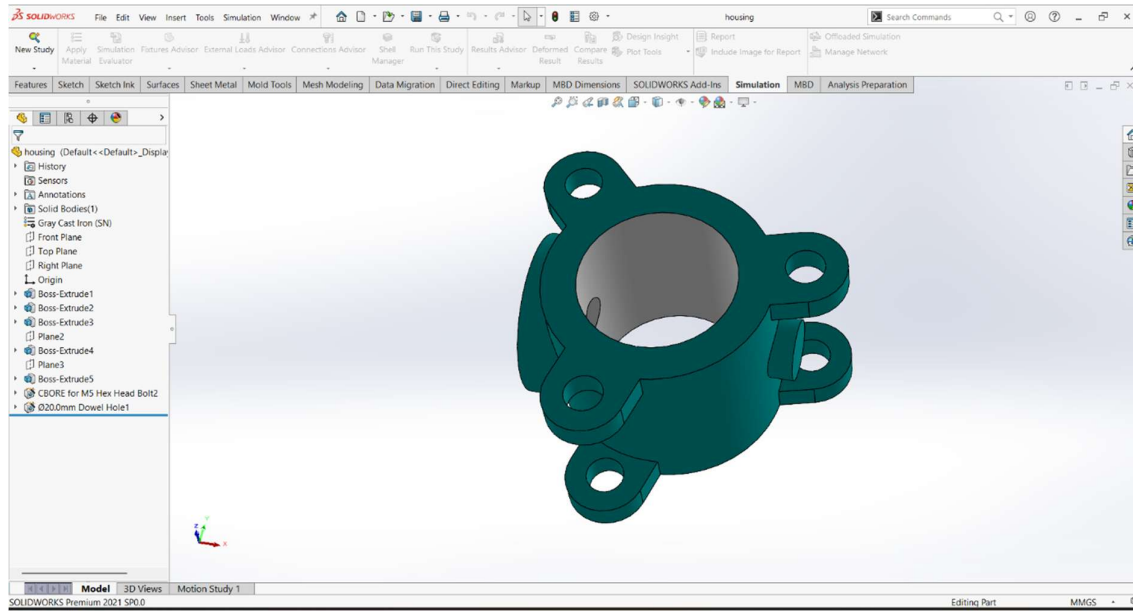


- Chamfer

- Hole Wizard



- Mirror

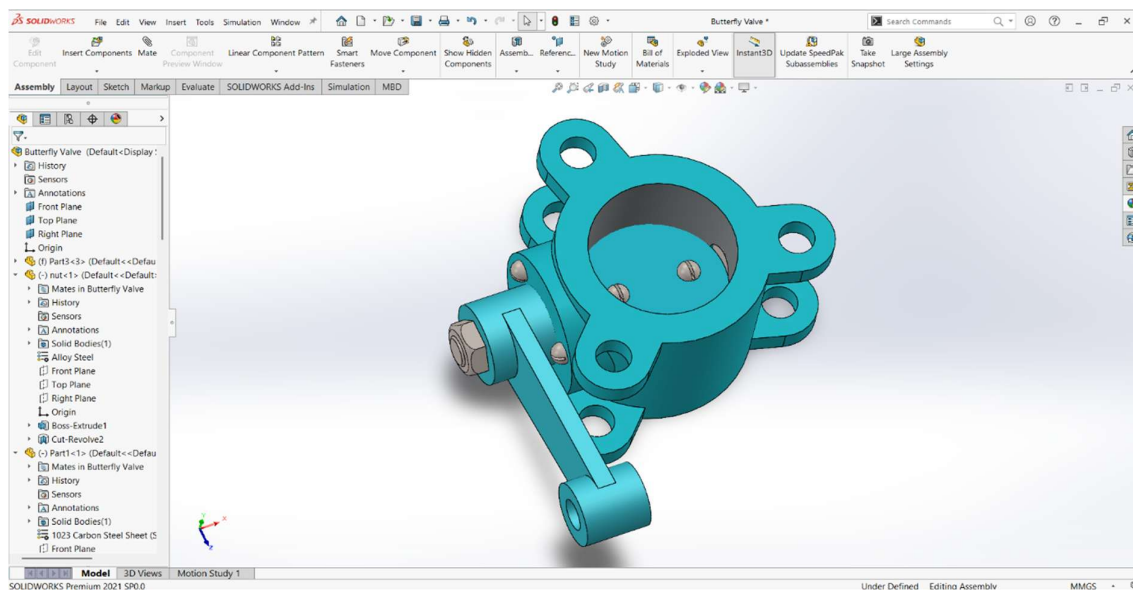


7. Assembly

After creating all individual parts, they were assembled using standard assembly mates.

The following mates were used:

- Coincident Mate
- Concentric Mate
- Distance Mate

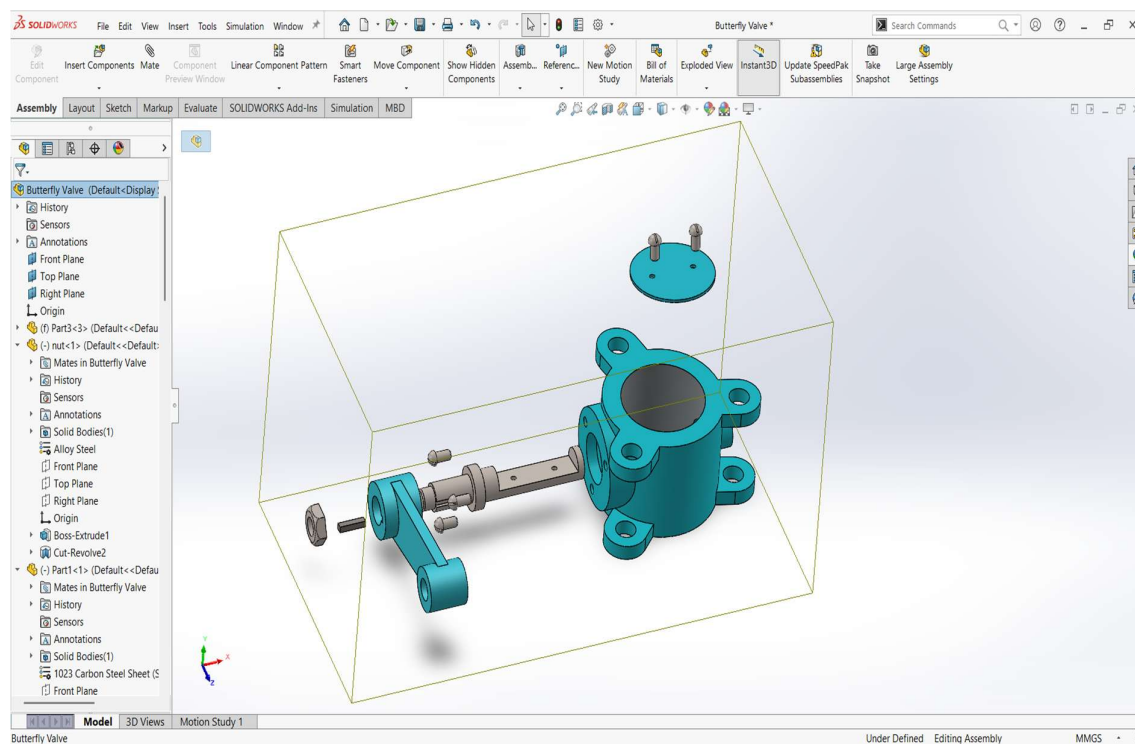


8. Exploded View

An exploded view was created to clearly demonstrate the position of every component before assembly.

This visualization helps understand:

- Assembly sequence
- Component arrangement
- Fastener locations
- Maintenance procedure

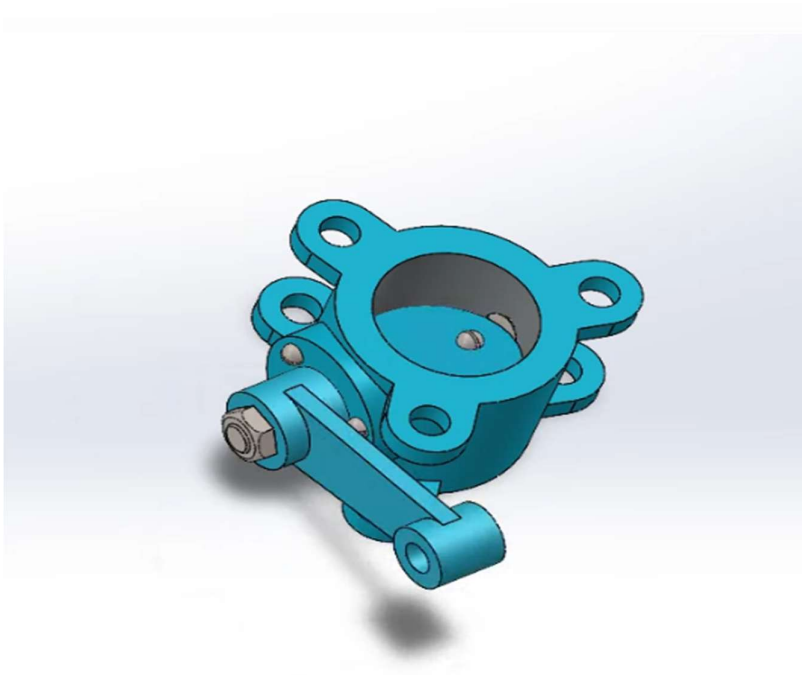
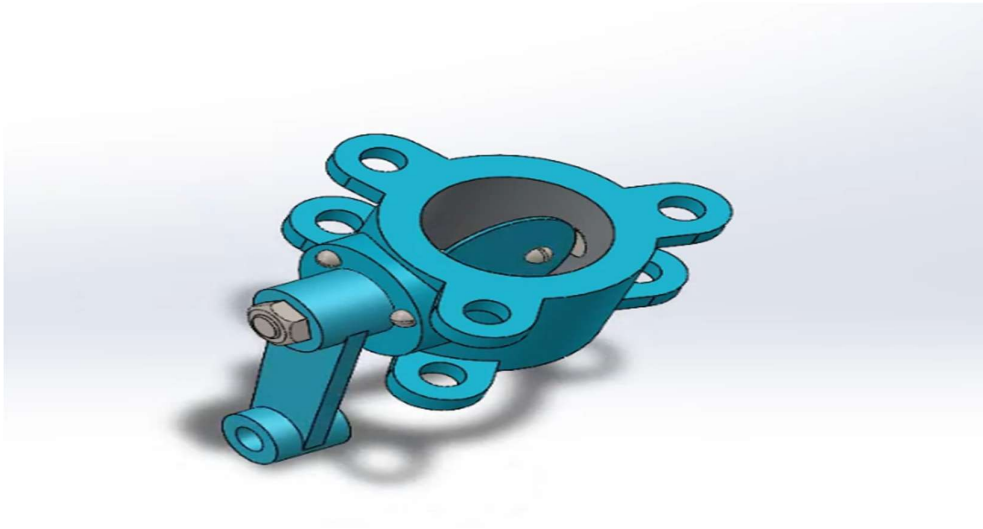


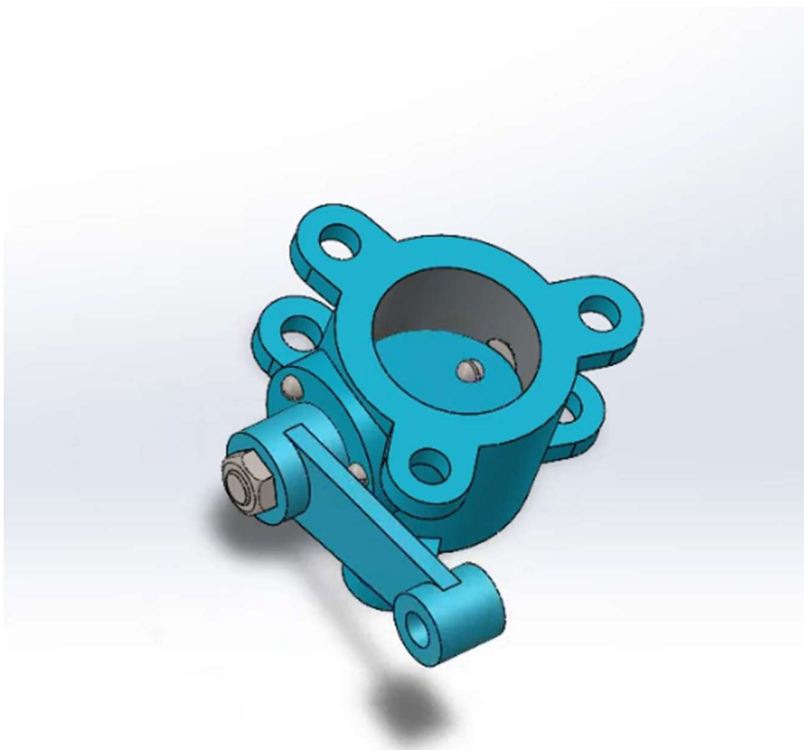
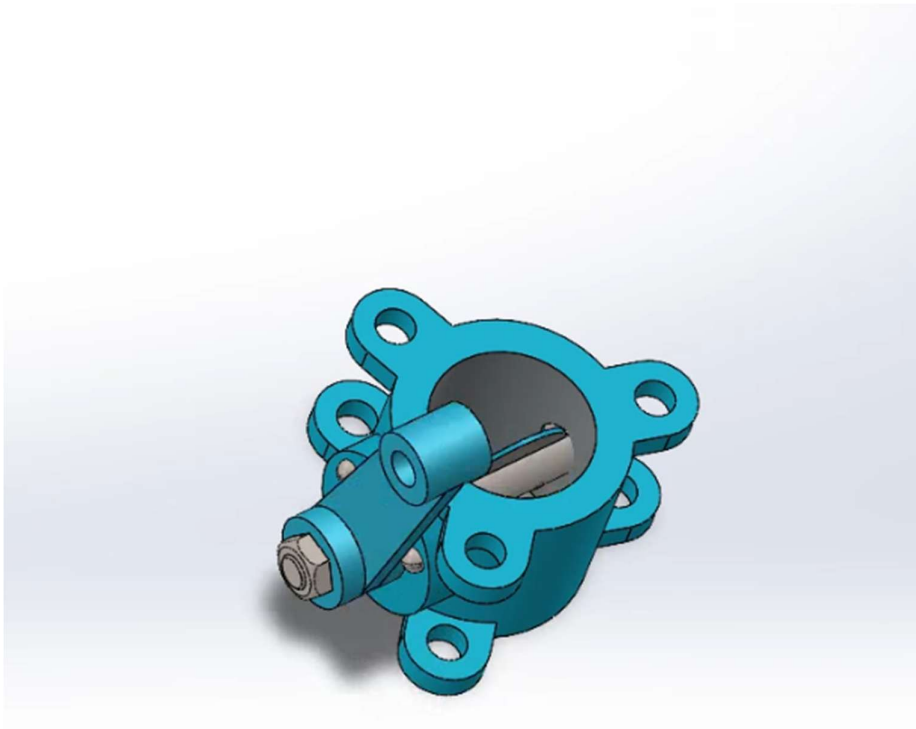
9. Motion Study

A motion study was developed to simulate the opening and closing operation of the butterfly valve.

The animation demonstrates the rotation of the lever and shaft, resulting in the movement of the valve disc.

This provides a better understanding of the working principle of the assembly.



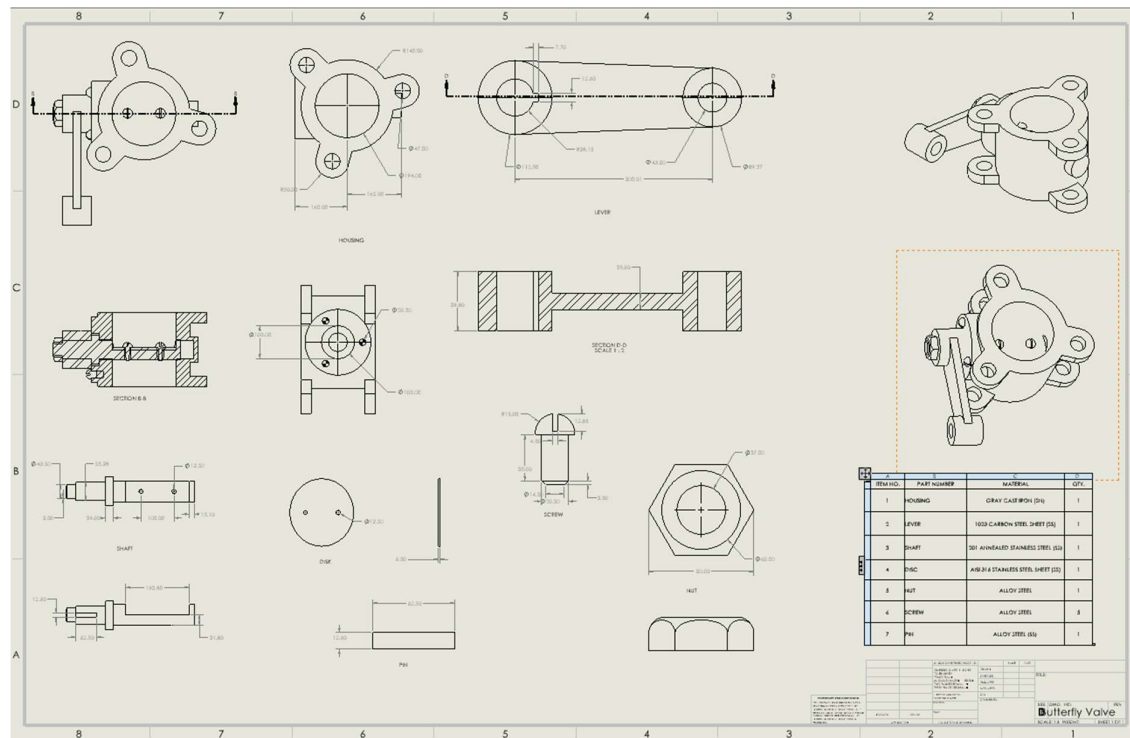


10. Engineering Drawings

Detailed engineering drawings were prepared for manufacturing purposes.

The drawing package includes:

- Orthographic Views
- Section Views
- Isometric Views
- Dimensions
- Bill of Materials (BOM)
- Material Information
- Part Detail



11. Manufacturing Process

The following manufacturing methods are suitable for producing the components:

Housing – Sand Casting followed by Machining

Disc – CNC Milling

Shaft – Turning and Grinding

Lever – Laser Cutting and Machining

Pin – Turning

Nut and Screws – Standard Fasteners

12. Design Challenges

During this project, I encountered several practical challenges:

- Maintaining correct assembly alignment
- Creating proper mating conditions
- Managing tolerances between components
- Preparing detailed engineering drawings
- Selecting suitable materials for different parts

12. Skills Demonstrated

Through this project, I demonstrated:

- 3D Part Modeling
- Assembly Design
- Engineering Drawings
- Exploded View Creation
- Motion Study
- Material Selection
- Bill of Materials (BOM)
- Design Documentation
- Mechanical Product Development

THANK YOU